The Research Lifecycle in Three Supply Chain Resilience Dissertations

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The research lifecycle runs from identifying a problem through design, data collection, analysis, interpretation, and dissemination, with ethical obligations carried through every stage. Material shortage is a recurring problem across very different supply chains, from engineered project supply chains (Kouvelis et al., 2023) to humanitarian relief networks (Singh et al., 2026), and it is the focus of my proposed study, *Mitigation of Supply Chain Uncertainties Caused by Material Shortage During Ramp-Up*, which examines semiconductor manufacturing equipment providers and their suppliers. This paper reviews three dissertations on supply chain disruption and resilience: a qualitative multiple case study (Benton, 2019), a quantitative survey study (Ponomarov, 2012), and a multiphase mixed methods study (Pettit, 2008), compared in Table 1 and Figure 1. A closing synthesis draws lessons for the proposed study.

Qualitative Dissertation: Benton (2019)

Research Problem, Purpose, and Questions

Benton (2019) addressed the problem that supply chain disruptions erode manufacturers' profitability and that some managers lack effective mitigation strategies. The purpose was to explore the strategies manufacturing managers use to mitigate disruptions to remain profitable, and the central research question asked what those strategies are. Consistent with exploratory qualitative inquiry, the study stated no hypotheses.

Methodology and Design

A qualitative methodology fit the goal of understanding how practitioners respond to disruptions rather than testing relationships among variables. The design was a multiple case study of two manufacturing firms in eastern Virginia, with four purposively selected supply chain managers who had successfully implemented mitigation strategies. Contingency theory of

fit, which links performance to aligning practices with environmental conditions, framed the study (Benton, 2019).

Research Instruments and Analysis

The researcher was the primary instrument. Data came from semistructured interviews guided by an interview protocol, company documents, and publicly available information, which allowed methodological triangulation. Benton (2019) used thematic analysis: transcripts and documents were coded, codes were compared across participants and firms until recurring themes emerged, and the themes were interpreted through contingency theory.

Key Findings

Three themes emerged: (a) collaboration and information sharing, (b) information technology and supply chain risk, and (c) the use of multiple suppliers (Benton, 2019). Managers reduced disruption risk by sharing information with partners, using technology to increase visibility, and avoiding single-source dependence.

Ethical and Legal Implications

Research with human participants requires institutional review board (IRB) approval, informed consent, and the right to withdraw. Small samples from identifiable firms risk reidentification, so pseudonyms and secure storage are essential. Legally, company documents may contain proprietary terms covered by nondisclosure agreements, which limits how they can be quoted.

Quantitative Dissertation: Ponomarov (2012)

Research Problem, Purpose, Questions, and Hypotheses

Ponomarov (2012) addressed a gap: resilience had been discussed conceptually, but little empirical evidence showed which firm capabilities produce it or whether it creates value. The

purpose was to examine the antecedents and value-based consequences of supply chain resilience at the firm level. The questions asked which orientations and capabilities lead to resilience and what outcomes it produces. Grounded in a dynamic capabilities extension of the resource-based view, the hypotheses linked supply chain orientation and risk management orientation to logistics capabilities, logistics capabilities to resilience, and resilience to firm-level consequences.

Methodology and Design

The methodology was quantitative and deductive, deriving hypotheses from theory and testing them statistically. The design was a cross-sectional, nonexperimental survey of supply chain and logistics managers (Ponomarov, 2012), estimating relationships among constructs across many firms.

Research Instruments and Analysis

The instrument was a structured questionnaire of multi-item scales measuring supply chain orientation, risk management orientation, logistics capabilities, and supply chain resilience. Ponomarov (2012) analyzed the data with structural equation modeling (SEM), first assessing the measurement model for reliability and convergent and discriminant validity, then estimating the structural model to test the hypothesized paths simultaneously.

Key Findings

The results supported a capability-based explanation of resilience. Firms' orientations toward supply chain and risk management shaped the logistics capabilities they developed, and those capabilities contributed to resilience, which carried value-based consequences for the firm (Ponomarov, 2012).

Ethical and Legal Implications

Survey research with professionals requires IRB review, informed consent, and respondent anonymity, particularly when questions touch on competitively sensitive practices. Researchers must also test for common method bias and report unsupported hypotheses transparently. Legally, electronically collected responses must comply with data privacy requirements.

Mixed Methods Dissertation: Pettit (2008)

Research Problem, Purpose, Questions, and Propositions

Pettit (2008) addressed the absence of an integrated, measurable framework for assessing and improving supply chain resilience. The purpose was to develop a conceptual framework, create and validate an assessment tool, and identify an implementation process. The research questions asked what constitutes resilience, whether it can be measured reliably, and which linkages between resilience factors should guide improvement. Instead of statistical hypotheses, the framework advanced propositions, notably that firms should seek a balanced zone in which capabilities match vulnerabilities without excessive cost.

Methodology and Design

The methodology was mixed methods with a multiphase sequential design (Pettit, 2008). Phase I built the framework from the literature and refined it with practitioner focus groups. Phase II validated an assessment tool by comparing quantitative scores with focus group data from global manufacturing and service firms that reviewed recent disruptions. Phase III identified critical linkages among resilience factors. Published supply chain studies show the same logic in concurrent form: Bimha et al. (2019) used a concurrent parallel design to study petroleum industry competitiveness, Asamoah et al. (2021) combined a survey, site observation,

and key informant interviews to assess vaccine cold chain management (see Figure 3), and Namisango et al. (2017) paired document review and group interviews with a key informant survey across three countries.

Research Instruments and Analysis

The instruments were focus group protocols and a structured self-assessment questionnaire on which respondents rated their firm's exposure on vulnerability attributes and its strength on capability attributes. Focus group data were coded and mapped to framework factors, and assessment scores were examined through correlations. Pettit (2008) then triangulated theoretical linkages, assessment correlations, and focus group connections, which is the defining integration step of mixed methods research.

Key Findings

Resilience can be assessed along two dimensions, vulnerabilities and capabilities. The study identified seven vulnerability factors comprising 40 attributes and 14 capability factors comprising 71 attributes, and triangulation produced 311 specific linkages that leaders can use to prioritize resilience investments (Pettit, 2008; see Figure 2).

Ethical and Legal Implications

Research inside corporations requires keeping firm identities and disruption histories confidential and reporting in aggregate. Focus groups limit confidentiality because participants hear one another, and university-industry collaboration requires managing conflicts of interest. Legally, the assessment tool later became a trademarked instrument, which raises intellectual property questions. Law can also shape the supply chain itself; Namisango et al. (2017) found that restrictive controlled-medicines legislation was a major barrier to opioid availability.

Comparative Synthesis and Implications for the Proposed Study

The research lifecycle changes with methodology. Benton (2019) built understanding inductively from a few information-rich participants, Ponomarov (2012) tested theory-derived hypotheses across a broader sample, and Pettit (2008) combined both logics. The findings converge on supplier collaboration, visibility, multiple sourcing, flexibility, and deliberate capability building. Modeling research points the same way: Kouvelis et al. (2023) show that inventories, time buffers, and supplier flexibility work together to manage material shortages, and Singh et al. (2026) build shortage mechanisms directly into supply chain optimization, which supports the analytics component of my proposed framework. The Appendix translates these findings into a risk-scoring algorithm with an illustrative example.

For the proposed qualitative case study, Benton's (2019) design offers a model for interviews, document review, and thematic analysis with equipment providers and suppliers, while Ponomarov's (2012) constructs and Pettit's (2008) vulnerability-capability lens can structure the coding of shortages, lead times, and shifts in supplier-buyer power. The semiconductor context sharpens the ethical and legal concerns, because firms operate under strict nondisclosure agreements and equipment data may be subject to export controls. IRB approval, informed consent, de-identification of firms, and careful screening of proprietary documents will therefore be central to the proposed research.

Table 1

Comparison of the Research Lifecycle Across the Three Dissertations

Element	Benton (2019)	Ponomarov (2012)	Pettit (2008)
Method	Qualitative	Quantitative	Mixed methods
Problem	Managers lack strategies to mitigate disruptions	Little evidence on what drives resilience	No integrated, measurable resilience framework
Questions / hypotheses	Central RQ on mitigation strategies; no hypotheses	Hypothesized paths: orientations → capabilities → resilience → value	Phase-based RQs; propositions on balanced resilience
Design	Multiple case study (2 firms, 4 managers)	Cross-sectional survey of SC and logistics managers	Multiphase sequential (3 phases)
Instruments	Interview protocol, company documents	Multi-item questionnaire scales	Focus group protocols, self-assessment tool
Analysis	Thematic analysis, triangulation	Structural equation modeling	Coding, correlations, triangulation
Key findings	Collaboration, IT and risk, multiple suppliers	Capabilities link orientations to resilience and value	7 vulnerability and 14 capability factors; 311 linkages
Ethics / legal	IRB, consent, confidentiality, NDAs	Anonymity, method bias, data privacy	Firm confidentiality, conflicts of interest, IP

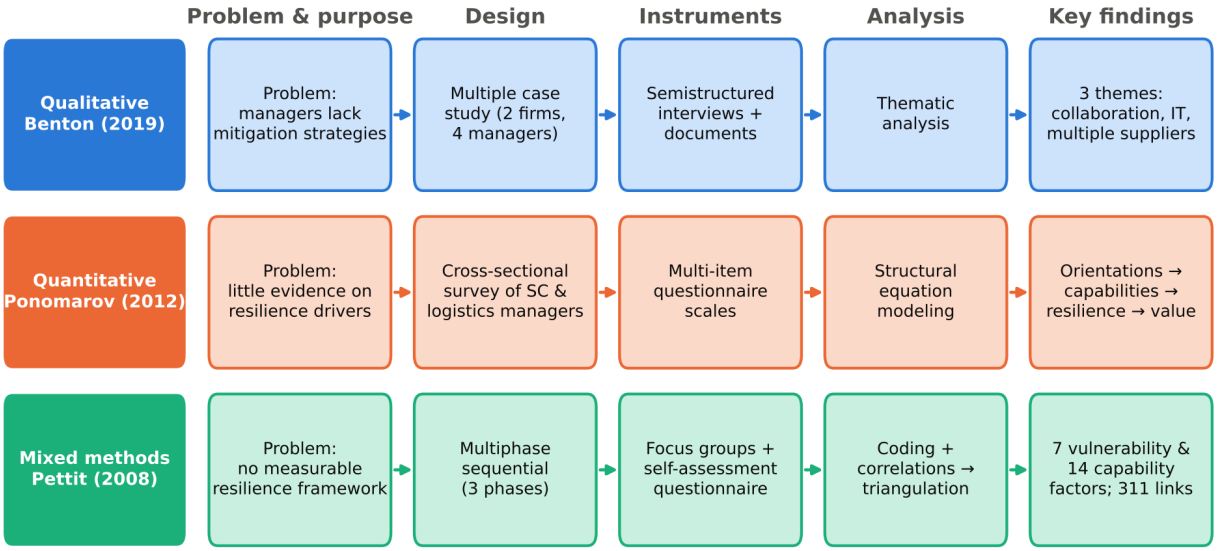
Note. Compiled by the author from Benton (2019), Ponomarov (2012), and Pettit (2008).

Column colors match Figure 1: blue = qualitative, orange = quantitative, green = mixed methods.

RQ = research question; SC = supply chain; NDA = nondisclosure agreement; IP = intellectual property.

Figure 1

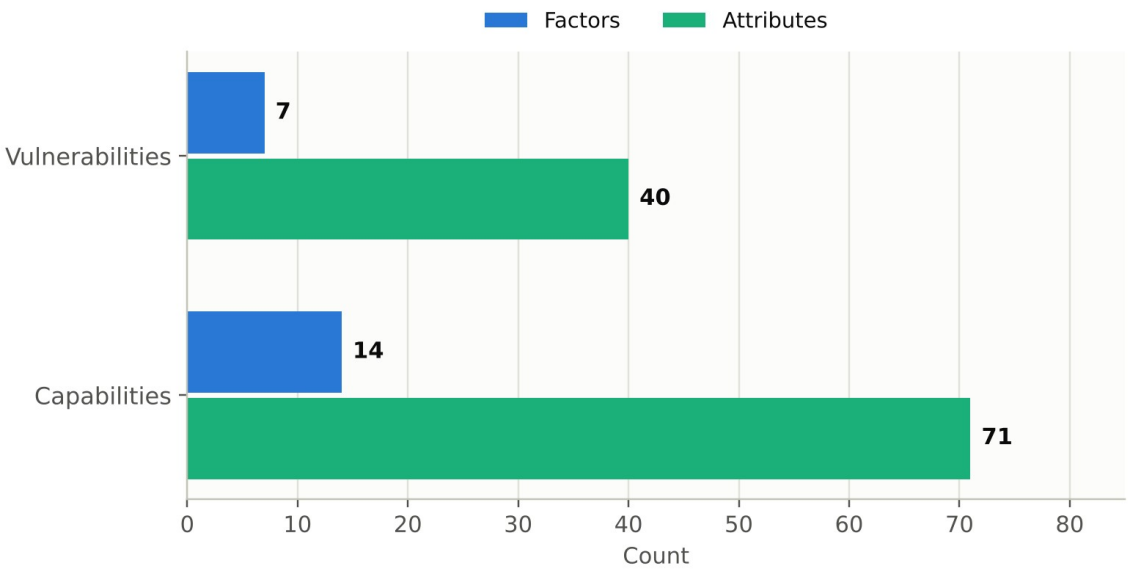
Research Lifecycle Stages in the Qualitative, Quantitative, and Mixed Methods Dissertations



Note. Each row traces one dissertation from problem to findings. Compiled from Benton (2019), Ponomarov (2012), and Pettit (2008).

Figure 2

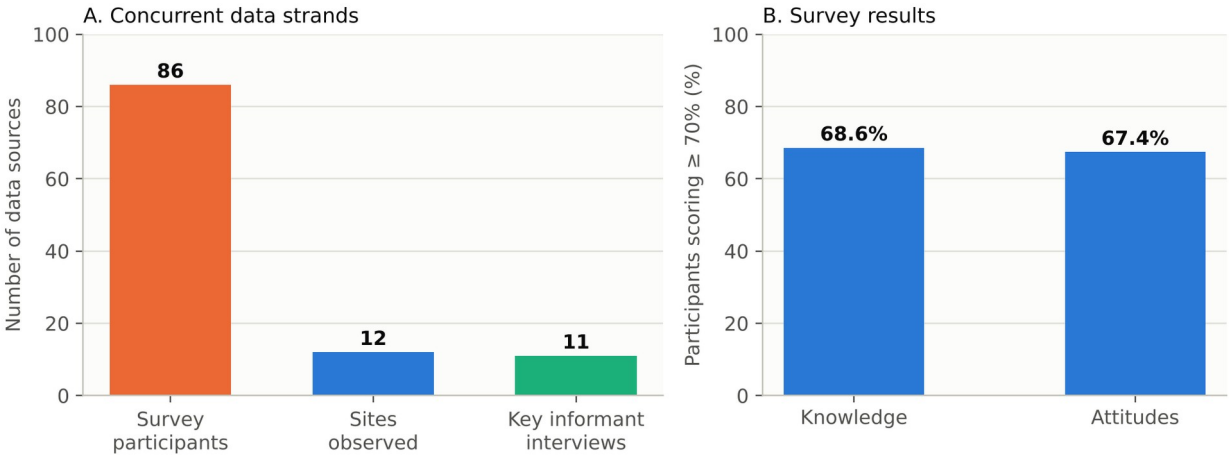
Structure of Pettit's (2008) Supply Chain Resilience Framework



Note. Counts of resilience factors and their underlying attributes. Triangulation of the two dimensions produced 311 vulnerability–capability linkages. Data from Pettit (2008).

Figure 3

Concurrent Data Strands and Survey Results in a Published Mixed Methods Supply Chain Study



Note. Panel A shows the three data sources collected concurrently across 12 facilities; Panel B shows the share of survey participants scoring at least 70% on cold chain knowledge and attitudes. Data from Asamoah et al. (2021).

Appendix

Material Shortage Risk-Scoring and Mitigation Selection Algorithm

Algorithm A1 operationalizes the proposed framework. It combines Pettit's (2008) vulnerability lens (coverage, lead time, source concentration, and supplier utilization) with the buffering levers that Kouvelis et al. (2023) identify (inventory, time buffers, and supplier flexibility) and the mitigation themes in Benton (2019). Figure A1 summarizes the steps, and Table A1 and Figure A2 apply the algorithm to illustrative data. The weights are starting assumptions to be calibrated with interview and performance data collected in the proposed study.

Algorithm A1

Material Shortage Risk Scoring (MSRS) for Ramp-Up Planning

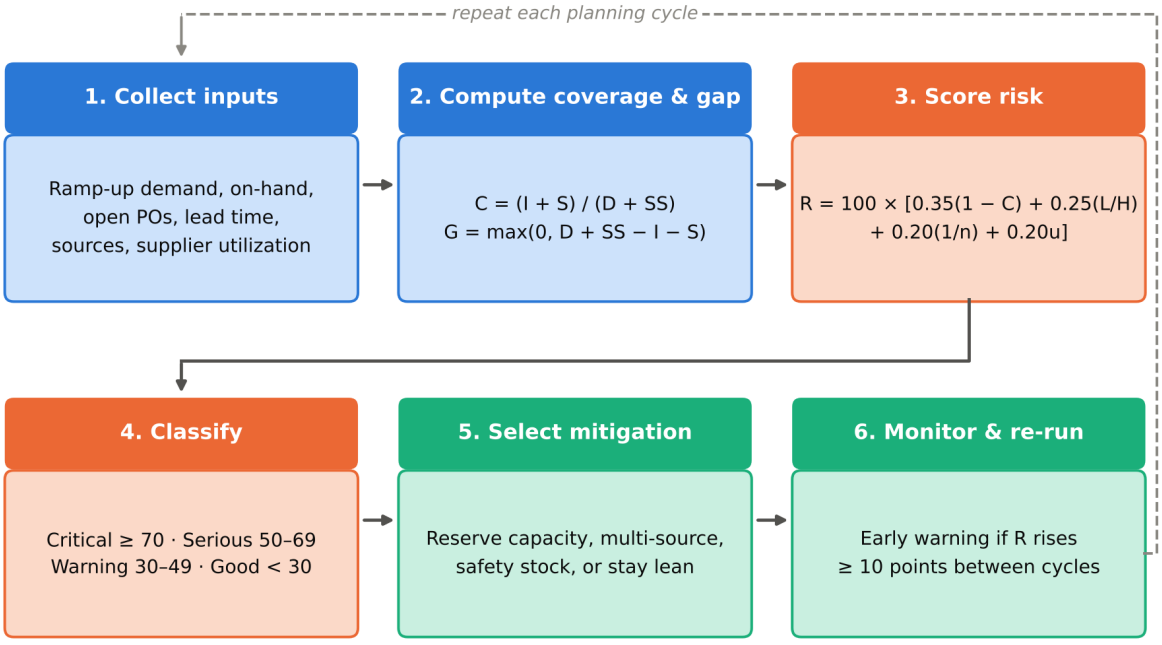
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Input:  for each critical part i in the ramp-up bill of materials:
        D_i  ramp-up demand over horizon H (weeks)
        I_i  on-hand inventory      S_i  confirmed open supply (POs)
        SS_i strategic safety stock L_i  supplier lead time (weeks)
        n_i  qualified sources      u_i  supplier capacity utilization (0-1)
        Weights w = (0.35, 0.25, 0.20, 0.20)
Output: risk score R_i, status, and mitigation action for every part

 1  for each part i do
 2      C_i <- (I_i + S_i) / (D_i + SS_i)                // projected coverage
 3      G_i <- max(0, D_i + SS_i - I_i - S_i)            // shortage gap (units)
 4      V_i <- w1(1 - min(C_i,1)) + w2 min(L_i/H,1)
              + w3(1/n_i) + w4 u_i                      // vulnerability index
 5      R_i <- round(100 * V_i)
 6      if   R_i >= 70 then status <- CRITICAL
 7           action <- reserve capacity, expedite, qualify alternate source
 8      elif R_i >= 50 then status <- SERIOUS
 9           action <- multi-source, raise strategic safety stock
10      elif R_i >= 30 then status <- WARNING
11           action <- weekly supplier capacity review, collaboration
12      else status <- GOOD;  action <- maintain lean / JIT replenishment
13      if R_i - R_i(previous cycle) >= 10 then raise EARLY WARNING
14  end for
15  sort parts by R_i (descending) and escalate CRITICAL parts to the
    cross-functional ramp-up review
16  repeat every planning cycle with refreshed demand and supplier data
```

Note. PO = purchase order; JIT = just-in-time. Developed by the author for the proposed study.

Figure A1

Flow of the Material Shortage Risk-Scoring Algorithm



Note. Blue steps prepare data, orange steps score and classify risk, and green steps act and monitor. C = coverage; G = gap; L/H = lead time relative to the planning horizon; n = qualified sources; u = supplier utilization.

Table A1

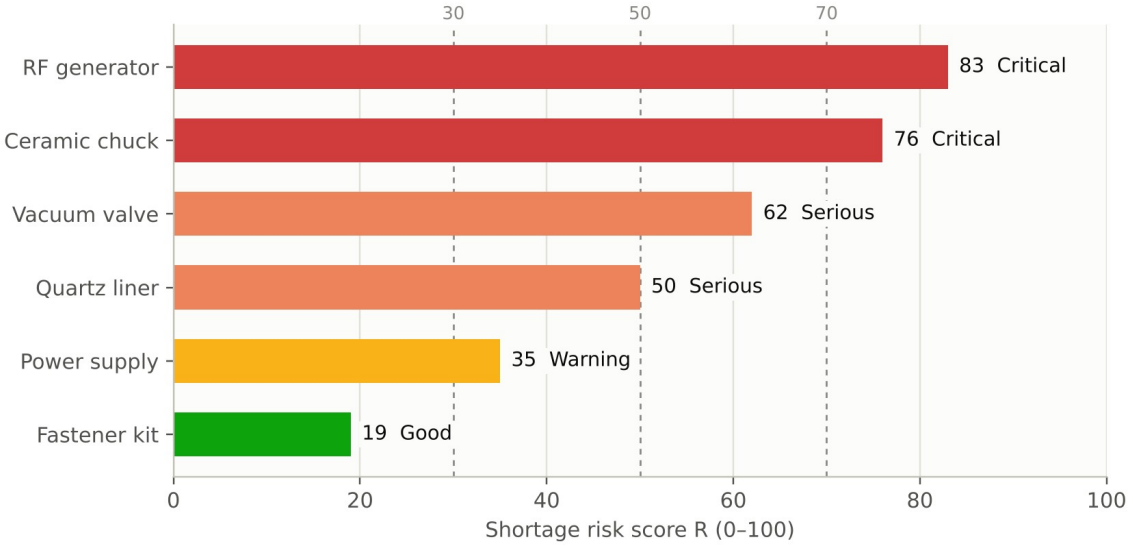
Illustrative Application of Algorithm A1 to Six Ramp-Up Parts (Hypothetical Data)

Part	D	I	S	SS	L	n	u	C	G	R	Status	Mitigation
RF generator	120	20	40	15	26	1	0.95	0.44	75	83	Critical	Capacity reservation, expedite, qualify alternate source
Ceramic chuck	90	15	45	10	20	1	0.85	0.60	40	76	Critical	Capacity reservation, expedite, qualify alternate source
Vacuum valve	300	80	150	30	14	2	0.80	0.70	100	62	Serious	Multi-source and raise strategic safety stock
Quartz liner	200	60	120	20	10	2	0.70	0.82	40	50	Serious	Multi-source and raise strategic safety stock
Power supply	150	70	90	15	8	3	0.60	0.97	5	35	Warning	Weekly supplier capacity review and collaboration
Fastener kit	800	600	400	50	3	4	0.40	1.18	0	19	Good	Maintain lean / just-in-time replenishment

Note. All values are hypothetical and were created only to demonstrate the calculation; they are not data from any firm or study. H = 13 weeks. D = ramp-up demand; I = on-hand; S = confirmed open supply; SS = safety stock; L = lead time in weeks; n = qualified sources; u = utilization; C = coverage; G = gap in units; R = risk score.

Figure A2

Illustrative Shortage Risk Scores by Part (Hypothetical Data)



Note. Dashed lines mark the status thresholds at 30, 50, and 70. Bars are colored by status (red = critical, coral = serious, amber = warning, green = good), and each bar is also labeled with its status so meaning does not depend on color alone. Values are computed from Table A1.

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